MATH1014 Calculus II Tutorial
Week 14

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1 Review

1.1 Vectors Basics (Magnitude and Unit Vectors)

A **vector** $\vec{v} = \langle v_1, v_2, v_3 \rangle$ has both direction and magnitude.

- **Magnitude (Length):** The length of $\vec{v}$ is $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$.
- **Unit Vector:** A vector of length 1. To find a unit vector $\vec{u}$ in the same direction as $\vec{v}$, divide $\vec{v}$ by its magnitude:
$$\vec{u} = \frac{\vec{v}}{|\vec{v}|}$$

1.2 Dot Product and Angle Between Vectors

The **dot product** of two vectors $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ is a scalar given by:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

- **Geometric Definition:** $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.
- **Finding the Angle:** You can solve for the angle using:
$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \implies \theta = \arccos\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right)$$

1.3 Cross Product and Orthogonal Vectors

The **cross product** $\vec{a} \times \vec{b}$ is a vector that is **orthogonal** (perpendicular) to both $\vec{a}$ and $\vec{b}$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$

1.4 Projections and Distance to a Plane

- **Vector Projection:** The projection of $\vec{b}$ onto $\vec{a}$ is the vector component of $\vec{b}$ in the direction of $\vec{a}$:
$$\text{Proj}_{\vec{a}}\, \vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2}\right) \vec{a}$$

- **Distance to a Plane:** If a plane is generated by two vectors $\vec{a}$ and $\vec{b}$ passing through the origin, the normal vector to the plane is $\vec{n} = \vec{a} \times \vec{b}$. The distance from a point R (with position vector $\vec{r}$) to this plane is the magnitude of the projection of $\vec{r}$ onto the normal vector $\vec{n}$:
$$d = |\text{Proj}_{\vec{n}}\, \vec{r}| = \frac{|\vec{r} \cdot \vec{n}|}{|\vec{n}|}$$

2 Exercises

Setup for Exercises: Given points $P = (-2, 1, 3)$, $Q = (-1, 0, 2)$, $R = (-3, 1, 0)$, and the origin $O = (0, 0, 0)$. Let's find the position vectors first:

$$\vec{OP} = \langle -2, 1, 3 \rangle, \quad \vec{OQ} = \langle -1, 0, 2 \rangle, \quad \vec{OR} = \langle -3, 1, 0 \rangle$$

2.1 Unit Vectors

Exercise 2.1. Find a unit vector in the same direction of $\vec{OP}$.

Solution. **Idea.** Calculate the magnitude of $\vec{OP}$ and divide each component by it.

First, the magnitude of $\vec{OP}$ is:

$$|\vec{OP}| = \sqrt{(-2)^2 + 1^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

The unit vector $\vec{u}$ is:

$$\vec{u} = \frac{\vec{OP}}{|\vec{OP}|} = \frac{1}{\sqrt{14}}\langle -2, 1, 3 \rangle = \left\langle -\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle = \left\langle -\frac{\sqrt{14}}{7}, \frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{14} \right\rangle$$

□

2.2 Dot Product and Angles

Exercise 2.2. Find the angle between $\vec{OQ}$ and $\vec{OR}$.

Solution. **Idea.** Use the geometric definition of the dot product to solve for θ.

First, compute the dot product:

$$\vec{OQ} \cdot \vec{OR} = (-1)(-3) + (0)(1) + (2)(0) = 3$$

Next, calculate the magnitudes:

$$|\vec{OQ}| = \sqrt{(-1)^2 + 0^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$$
$$|\vec{OR}| = \sqrt{(-3)^2 + 1^2 + 0^2} = \sqrt{9 + 1} = \sqrt{10}$$

Using the formula $\cos\theta = \frac{\vec{OQ} \cdot \vec{OR}}{|\vec{OQ}||\vec{OR}|}$:

$$\cos\theta = \frac{3}{\sqrt{5}\sqrt{10}} = \frac{3}{\sqrt{50}} = \frac{3}{5\sqrt{2}} = \frac{3\sqrt{2}}{10}$$

Therefore, the angle is:

$$\theta = \arccos\left(\frac{3\sqrt{2}}{10}\right)$$

□

2.3 Cross Product and Orthogonality

Exercise 2.3. Find a vector orthogonal to both $\vec{OP}$ and $\vec{OQ}$.

Solution. **Idea.** The cross product of two vectors is orthogonal to both of them.

We compute $\vec{OP} \times \vec{OQ}$:

$$\begin{aligned}
\vec{OP} \times \vec{OQ} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & 3 \\ -1 & 0 & 2 \end{vmatrix} \\
&= \begin{vmatrix} 1 & 3 \\ 0 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -2 & 3 \\ -1 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -2 & 1 \\ -1 & 0 \end{vmatrix} \mathbf{k} \\
&= (1(2) - 3(0))\mathbf{i} - (-2(2) - 3(-1))\mathbf{j} + (-2(0) - 1(-1))\mathbf{k} \\
&= 2\mathbf{i} - (-4 + 3)\mathbf{j} + (0 + 1)\mathbf{k} \\
&= 2\mathbf{i} + 1\mathbf{j} + 1\mathbf{k} = \langle 2, 1, 1 \rangle
\end{aligned}$$

So, $\vec{n} = \langle 2, 1, 1 \rangle$ is a vector orthogonal to both $\vec{OP}$ and $\vec{OQ}$. □

2.4 Projections and Distances

Exercise 2.4. Find the projection of $\vec{OR}$ on the direction orthogonal to both $\vec{OP}$ and $\vec{OQ}$.

Solution. **Idea.** We need the vector projection of $\vec{OR}$ onto the normal vector $\vec{n} = \vec{OP} \times \vec{OQ}$ we found in the previous exercise.

From the previous exercise, $\vec{n} = \langle 2, 1, 1 \rangle$. First, calculate the dot product $\vec{OR} \cdot \vec{n}$:

$$\vec{OR} \cdot \vec{n} = \langle -3, 1, 0 \rangle \cdot \langle 2, 1, 1 \rangle = -3(2) + 1(1) + 0(1) = -6 + 1 = -5$$

Next, calculate the squared magnitude of $\vec{n}$:

$$|\vec{n}|^2 = 2^2 + 1^2 + 1^2 = 4 + 1 + 1 = 6$$

Now, compute the vector projection:

$$\text{Proj}_{\vec{n}} \vec{OR} = \left(\frac{\vec{OR} \cdot \vec{n}}{|\vec{n}|^2} \right) \vec{n} = \frac{-5}{6} \langle 2, 1, 1 \rangle = \left\langle -\frac{10}{6}, -\frac{5}{6}, -\frac{5}{6} \right\rangle = \left\langle -\frac{5}{3}, -\frac{5}{6}, -\frac{5}{6} \right\rangle$$

□

Exercise 2.5. Find the distance of R to the plane generated by $\vec{OP}$ and $\vec{OQ}$.

Solution. **Idea.** The distance from a point to a plane passing through the origin is the magnitude of the projection of the point's position vector onto the plane's normal vector.

The normal vector to the plane generated by $\vec{OP}$ and $\vec{OQ}$ is exactly $\vec{n} = \vec{OP} \times \vec{OQ}$. We already calculated the projection of $\vec{OR}$ onto this normal vector in the previous exercise:

$$\text{Proj}_{\vec{n}} \vec{OR} = \left\langle -\frac{5}{3}, -\frac{5}{6}, -\frac{5}{6} \right\rangle$$

The distance is the magnitude of this projection vector:

$$\begin{aligned}
d &= \left|\left\langle -\frac{5}{3}, -\frac{5}{6}, -\frac{5}{6}\right\rangle\right| = \sqrt{\left(-\frac{5}{3}\right)^2 + \left(-\frac{5}{6}\right)^2 + \left(-\frac{5}{6}\right)^2} \\
&= \sqrt{\frac{25}{9} + \frac{25}{36} + \frac{25}{36}} = \sqrt{\frac{100}{36} + \frac{25}{36} + \frac{25}{36}} = \sqrt{\frac{150}{36}} \\
&= \frac{\sqrt{150}}{6} = \frac{\sqrt{25 \cdot 6}}{6} = \frac{5\sqrt{6}}{6}
\end{aligned}$$

$\square$